College for Professional Studies
3333 Regis Boulevard
Denver, CO 80221

Regis University
College for Professional Studies
School of Computer & Information Sciences

SYLLABUS

COURSE TITLE: Assembly Programming

COURSE NUMBER: CS 370

Course Description: Describes the elements and techniques of assembly language programming for microprocessors used in the IBM compatible family of microcomputers. Introduces computer architectures, and discusses the concepts of data representations, processing instructions, addressing modes, macros, functions and procedures and file I/O.

Course Prerequisites

CS208 Computer Science Fundamentals
CS362 Data Structures

In order to successfully complete this course, students are expected to have taken the prerequisite CS362 course or its equivalent. From CS362, you should have a working knowledge of the following topics:

1. Practice with problem definition, solution construction and algorithmic development using top-down design techniques.
2. Ability to implement control structures used for sequencing, selection, and iteration.
3. Ability to choose and implement appropriate data structures including arrays, structures, linked lists, and file processing.
4. Understanding of modular code design and how to design test cases.

Although CS370 introduces Assembly Programming, it is also a Computer Science course that introduces advanced programming concepts. Consequently, the ability to program at the CS362 learning outcome level is essential for successful in this course.

CS370 is challenging and covers a substantial amount of material at a rapid pace. Ultimately, programming is a skill that requires the ability to put designs into practice. The only way to succeed is to practice this skill. Therefore, this course will require a **significant** amount of time each week to complete the programming assignments. If you are not willing or able to spend the necessary time, please reconsider whether this is the correct time to attend this class.

Course Outcomes

Upon successful completion of this course, students should be able to:

- Describe how the fetch-execute cycle is used during runtime in a classical von Neumann machine.
- Create programs using various instructional formats such as addresses per assembly language command, fixed versus variable length formats, and data addressing modes.
- Compare and contrast Assembly language constructs with prior learning about high-level programming language (e.g., C++) constructs.
- Create programs that perform subroutine calls using the Assembly programming language.
- Explain how subroutine calls are handled at the assembly level.
- Master how numeric and non-numeric data are represented in memory including converting numerical data from one base to another such as hexadecimal, binary, and decimal.
- Design, develop, execute, and debug assembly language programs (including string manipulation, subroutine calls, and representing arrays).
- Create and execute test plans to verify program code works correctly.

Course Materials and Resources, including Technology Tools

Required Textbook:

Irvine, K. R. (2011). *Assembly language for x86 processors* (6th ed.). Upper Saddle River, NJ: Prentice Hall (Pearson Education). ISBN13: 978-0-13-602212-1 or ISBN10: 0-13-602212-X hardcover.

Additional Required Resources:

Online shell available via the Regis Worldclass learning environment: http://online.regis.edu

Technology Tools:

Visual Studio 2010 Express containing Microsoft Assembler (MASM), which can be downloaded (free) from the textbook author's Website: http://www.kipirvine.com/asm/ (select the "Getting started with MASM" link).

Pre-Assignment

- Complete assigned readings in the Course Assignments and Activities table below
- Be prepared to ask questions on unclear areas in the assigned reading
- Familiarize yourself with the procedures for accessing the MASM assembler (above)

Related Standards

Related Standards	
Program Outcomes	
	This program enables students to achieve the following program outcomes by the time of graduation:
	An ability to apply knowledge of computing and mathematics appropriate to the discipline
	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution
	An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs
	An ability to communicate effectively with a range of audiences
	An ability to use current techniques, skills, and tools necessary for computing practice
	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices
	An ability to apply design and development principles in the construction of software systems of varying complexity
Regis Nine Educational Outcomes	
	Content Knowledge in Discipline
	Diverse Cultures, Perspectives, and Belief Systems
	Arts, Sciences, and Humanities
	Critical Thinking
	Communication
	Technology
	Ethical and Social Responsibility
	Leadership in Service to Others
	Lifelong Learning
Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities	
	Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives in Action, Care of the Whole Person, Men and Women for Others

Course Assignment and Activities

Week	Learning Topics	Reading Assignments	Graded Assignments and Associated Course Points*
1	Computer Number Systems and Architectures Fundamental Assembly Language Elements: data definition and data movement instructions	Textbook: Chapters 1, 2, 3, & Chapter 4 §4.1 – 4.1.10 Online Regis Worldclass	Introductions Pretest Quiz Weekly Participation (1.25%) Programming Assignment 1 (10%)
2	Basic Arithmetic and Loop Instructions	Textbook: Sec. 3.4-3.7 & Sec 4.1-4.1.10 Online Regis Worldclass	Discussion Questions Programming Assignment 2 (10%) Jesuit Key Values Essay
3	Procedures and the Stack and Conditional Processing Instructions	Textbook: Rest of Ch. 4 Online Regis Worldclass	Weekly Participation Programming Assignment 3(10%)
4	Boolean Instructions and Implementing Control Structures in Assembly Language	Textbook:Ch. 5 Online Regis Worldclass	Weekly Participation Programming Assignment 4 (10%) Midterm (10%)
5	Shift and Rotate Instructions	Textbook: Ch. 6 Online Regis Worldclass	Weekly Participation Programming Assignment 5 (Part I: Documentation)
6	Parsing Integer Data, and Extended Precision Arithmetic	Textbook: Ch. 7 Online Regis Worldclass	Weekly Participation Programming Assignment 5 (10%) (Part II: Program)
7	Procedures and Parameters, Stack Frames, Multiple File Programs	Textbook: Ch. 8 Online Regis Worldclass	Weekly Participation Programming Assignment 6 (Part I: Documentation) Discussion Questions
8	String Instructions, String Representations and Array Representations and Processing	Textbook: Ch. 9. Online Regis Worldclass	Weekly Participation Programming Assignment 6 (10%) (Part II: Program) Final exam (15%)

* Note to Classroom sections only: your facilitator will give you the exact due dates.
Online due dates are Sunday, Midnight Mountain Time.

Programming Assignments

Each programming assignment will involve designing, implementing, and testing a program using the concepts discussed in the book and class. **Note:** programs that do not assemble, are not modular, nor properly documented standards will not be accepted.

Programming assignments will be graded according to details listed in the rubrics given in the class, which also includes the following criteria:

1. Source Code
   a. Code is easy to read and self-documenting
   b. Appropriate comments are used
2. Coding Style
   a. When applicable, the program is modular
   b. The Java programming language is used correctly
3. Functionality
   a. There are no assembly, logic, and design errors.
   b. The program solves the assigned problem.
4. Testing, as appropriate
5. Analysis (if required)
   a. Logical and complete

Exams

There will be a midterm and final exam. Exams questions will be cumulative, taken from reading assignments, programming assignments, and class participation. Exams may test your understanding, comprehension, application, and evaluation of the material presented in this class.

Participation

Class participation/effort is important because we can all learn from each other. Your participation points can make a difference in the final grade. Participation means:

1. Present in class every session (classroom)
   Present in the forum every week (online)
2. Effectively responds to questions from the facilitator (classroom)
   Regularly checks forum and posts all required items by the deadlines (online)
3. Interacts/replies to other students in classroom/forum discussions.

COLLEGE FOR PROFESSIONAL STUDIES POLICIES (LINKS):

You will be asked to enter a valid Regis University UserID and password to access the following policies.

Academic Integrity
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/Academic%20Integrity%20Policy%20and%20Board%20-%20December%2020,%202010.doc

In cases of suspected cheating or plagiarism, the instructor will discuss the matter with the student(s) involved. If necessary, the instructor reserves the right to question any student orally or in writing and to use my evaluation of the student's understanding of the assignment and of the submitted solution as evidence of cheating.

Attendance Participation
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Attendance%20Participation.pdf

Confidential Proprietary Information Policy
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Confidential%20Proprietary%20Information%20Policy.pdf

Dayton Memorial Library
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Dayton%20Memorial%20Library.pdf

Diversity
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Diversity%20at%20RU.pdf

Equal Access and Disability Services
https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Equal%20Access_Disability%20Services.pdf

Human Subjects Review (IRB)
http://www.regis.edu/Academics/Academic-Research-and-Grants/Regis%20Review%20Boards.aspx#.UUtE8WdYkdU

Inclement Weather and Class Cancellation
In the event of inclement weather (i.e., blizzard), students should call the University Weather line at 303-458-1818 for closure information.

Late or Incomplete Course Procedures

https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Late%20Assignments_Assignment%20Revisions%20Policy%20CPS.pdf

Late Policy for CS434

A deduction of 3% per day up to one week late, at which point the assignment is no longer accepted. Week 8 assignments and exams cannot be submitted late. ONLINE: Late discussion posts will have 10% deducted for each day the post is late. Discussion posts that are posted more than ONE week after the official due date will not be given ANY points.

Incomplete Policy for CS434

In general, incompletes will not be granted unless you've completed at 6 weeks of the course.

Learner Conduct

https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Learner%20Conduct.pdf

Writing Assistance

https://in2.regis.edu/sites/spsdean/CPS%20Approved%20Policies%20and%20Procedures/CPS%20Syllabus%20Policies/Writing%20Assistance.pdf

OTHER INFORMATION:

NOTE TO LEARNERS: On occasion, the course faculty may, at his or her discretion, alter the Learning Activities shown in this Syllabus. The alteration of Learning Activities may not, in any way, change the Learner Outcomes or the grading scale for this course as contained in this syllabus. Examples of circumstances that could justify alterations in Learning Activities could include number of learners in the course; compelling current events; special faculty experience or expertise; or unanticipated disruptions to class session schedule.